Project Proposal: From Accidents to Weapons

A Statistical Analysis of AI Incidents Using the AI Incident Database

1. Project Metadata

Attribute	Details
Student Name	Poushikan R
Student ID	2301107074
Section / Cohort	B
Data Source	AI Incident Database

2. Project Overview

This project presents a comprehensive analysis of 1,711 reported Artificial Intelligence incidents spanning from 1983 to 2026. **Initially**, the data is sourced directly from the AI Incident Database, **after which** it is processed utilizing Python for advanced data evaluation.

3. Research Objectives & Scope

Specifically, the research examines how the frequency of these AI-related incidents has escalated over time. **Furthermore**, it rigorously tests the hypothesis of whether AI-related harm is shifting from accidental system failures toward deliberate misuse by human actors.

4. Analytical Methodology

To achieve these research objectives, the methodology employs descriptive statistics alongside trend regression. **Additionally**, chi-square tests are utilized to effectively compare incident distributions across various years, risk domains, and underlying intents.
Moreover, advanced Natural Language Processing techniques—notably TF-IDF combined with K-Means clustering—are applied to group the textual incident descriptions. **Consequently**, this crucial step helps determine whether the emergent text patterns accurately align with the predefined and labeled risk categories.

5. Technical Stack

Finally, the entire analytical pipeline is powered by a robust stack of Python libraries, ensuring rigorous statistical analysis and clear visualization of the findings. The primary libraries used include:

- **Pandas & NumPy:** Utilized for comprehensive data manipulation and numerical operations.
- **SciPy & Scikit-learn:** Applied for advanced statistical testing and machine learning clustering.
- **Matplotlib:** Deployed for generating visual trends and data distributions.